

Review retrospective:
*the harmonic Gibbs moments — and a dead block that was
load-bearing for the linter*

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Abstract

A soundness review of `Laplace/OneD/Harmonic.lean`: the harmonic Gibbs measure — partition function $Z_\lambda(t) = \sqrt{2\pi/(\lambda t)}$, the moments $\langle x^{2k} \rangle_t = (2k-1)!!/(\lambda t)^k$ and $\langle x^{2k+1} \rangle_t = 0$, and $\text{Cov}_t[x^2, x] = 0$. Result: **a clean fidelity pass with one landed edit (a dead open Set) and one instructive non-edit** — a mathematically-dead proof block that turned out to be load-bearing for the `unusedVariables` linter, reverted once the rebuild exposed it.

1 Setting

The harmonic case is the leading-order Laplace computation and the base the anharmonic programme perturbs around (`harmonicPotential` is what the anharmonic base decomposes against). Its moments reduce directly to the Gaussian-moment lemmas by setting the inverse covariance to λt .

2 What was reviewed

A definition, a private integrand-rewrite helper, and six theorems: the even/odd numerator integrals, the partition function, the even/odd Gibbs expectations, and the harmonic covariance $\text{Cov}_t[x^2, x] = 0$.

3 Fidelity / soundness findings

Sound; all six theorems check out. $Z = \sqrt{2\pi/(\lambda t)}$ is the $k = 0$ even-moment value $((2 \cdot 0 - 1)!! = 0!! = 1, (\lambda t)^{-1/2})$; the even expectation is the quotient $(\lambda t)^{-(k+1/2)}/(\lambda t)^{-1/2} = (\lambda t)^{-k}$; the odd expectations vanish by symmetry; and $\text{Cov}_t[x^2, x] = 0$ because $x^2 \cdot x = x^3$ and x are both odd powers — the $\alpha = 0$ specialisation of the primer's $\text{Cov}_t[x^2, x] = -2\alpha/(\lambda^3 t^2) + o(t^{-2})$.

4 Simplifications applied — and one revealing revert

Landed: a dead open Set. The file opened `Real MeasureTheory Set`, but no `Set` name is used (the integrals are over all of $\mathbb{R}$ with `volume`; there is no `IoI` or interval). Removed; `Real` (bare π), `MeasureTheory` ($\int/\text{volume}$), and `open scoped Nat (!)` all stay. Rebuild green, no warnings, eight declarations byte-identical.

Reverted: a linter-load-bearing dead block. GPT also flagged — correctly, in the mathematical sense — that `harmonic_int_pow_odd` contains a change-of-variables detour (`have this : ... := ...; clear this`) that is computed and then immediately `cleared`, the actual proof going through the odd-function symmetry `integral_neg_eq_self`. The block is dead. But removing it made the `unusedVariables` linter fire on the theorem’s *public* hypotheses `hlam : 0 < lam` and `ht : 0 < t`: the only thing consuming them in that proof was the detour’s `have hbpos := mul_pos hlam ht`; the symmetry argument needs no positivity. Because `hlam` and `ht` are part of the frozen public signature (they mirror the even-moment theorem’s interface), they cannot be renamed to `_hlam`/`_ht` or dropped, and leaving them produces new warnings. So the “dead” block is the mechanism that keeps the public hypotheses syntactically used — it is load-bearing for the linter. The rebuild caught it (green build, but two new `unusedVariables` warnings), and the removal was reverted.

GPT’s remaining suggestions (the x^{2-0} rewrite, the `gibbsCov` function-equality `shows`, factoring the duplicated $\sqrt{\text{rpow}}$ algebra, the universal $(abc)/(bd)$ rewrite) were declined as tactic-golf.

5 Disagreements and open questions

None. GPT affirmed both the landed removal and the revert.

6 Lessons

- **A mathematically-dead block can be load-bearing for the `unusedVariables` linter under a frozen signature.** When a proof carries public hypotheses it does not truly need (here, positivity that a symmetry argument ignores), a “dead” detour may be the only thing consuming them. Remove it and the linter flags the public hypotheses — which a review tide cannot rename or drop. This is the seabed’s “dead code feeds automation” gotcha in a sharper form: it is not a local that feeds a later tactic, but a block that feeds the *linter* on behalf of the interface. The rebuild-plus-linter gate is exactly what catches it; “the build is green” is necessary but not sufficient — “no new warnings” is the real bar.
- **Two dead-code flags in one file can have opposite verdicts.** The dead `open` was real and safe; the dead block was real but unsafe to remove. Triage each against the rebuild independently; never batch dead-code removals on the assumption they are all equally safe.

7 Follow-ups

- If the redundant positivity hypotheses on `harmonic_int_pow_odd`/`_pow_odd` expectations are ever to be cleaned up, that is a forward tide (a deliberate signature change weakening the interface), not a review-tide edit.
- Remaining tractable 1D never-reviewed files: `Localisation.lean`, the J-function and `Quartic/Sextic/Monomial` families.